

Butyrate and other short-chain fatty acids as modulators of immunity: what relevance for health?



PURPOSE OF REVIEW: High-fiber diets have been shown to reduce plasma concentrations of inflammation markers. Increased production of fermentation-derived short-chain fatty acids (SCFAs) is one of the factors that could exert these positive effects. This review examines the effects of SCFAs on immune cells and discusses the relevance of their effects on systemic inflammation, as frequently seen in obesity. RECENT FINDINGS: SCFAs have been shown to reduce chemotaxis and cell adhesion; this effect is dependent on type and concentration of SCFA. In spite of conflicting results, especially butyrate seems to have an anti-inflammatory effect, mediated by signaling pathways like nuclear factor- κ B and inhibition of histone deacetylase. The discrepancies in the results could be explained by differences in cell types used and their proliferative and differentiation status. SUMMARY: SCFAs show anti-inflammatory effects and seem to have the potency to prevent infiltration of immune cells from the bloodstream in, for example, the adipose tissue. In addition, their ability to inhibit the proliferation and activation of T cells and to prevent adhesion of antigen-presenting cells could be important as it recently has been shown that obesity-associated inflammation might be antigen-dependent. More studies with concentrations in micromolar range are needed to approach more physiological concentrations.